

# CHE 201 Spring 2020 HW 4

Thanh Nguyen

TOTAL POINTS

**97 / 100**

## QUESTION 1

### 1 Question 1 20 / 20

✓ - **0 pts** Correct

- **5 pts** "...in terms of their individual thermal efficiencies"

- **2.5 pts** Insufficient component about the phase-out schedule

- **5 pts** Insufficient component about the replacer

- **2.5 pts** Insufficient component about the replacer

- **1 pts** No enough references

## QUESTION 2

### 2 Question 2 20 / 20

✓ - **0 pts** Correct

- **5 pts** Wrong V2

- **5 pts** Wrong work

- **5 pts** Wrong internal energy difference

- **5 pts** Wrong final temperature

- **20 pts** Did not answer 2.53

## QUESTION 3

### 3 Question 3 20 / 20

✓ - **0 pts** Correct

## QUESTION 4

### 4 Question 4 17 / 20

- **0 pts** Correct

- **20 pts** Did not answer problem 3.7

✓ - **3 pts** (a) Wrong sketch of T-v diagram

- **3 pts** (a) Wrong specific volume

- **3 pts** (b) Wrong sketch of T-v diagram

- **3 pts** (b) Wrong specific volume

- **3 pts** (c) Wrong sketch of T-v diagram

- **3 pts** (c) Wrong specific volume

## QUESTION 5

### 5 Question 5 20 / 20

✓ - **0 pts** Correct

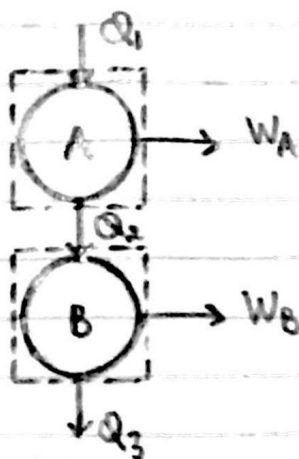
- **20 pts** Did not answer 3.4D

- **5 pts** Insufficient component about the phase-out schedule

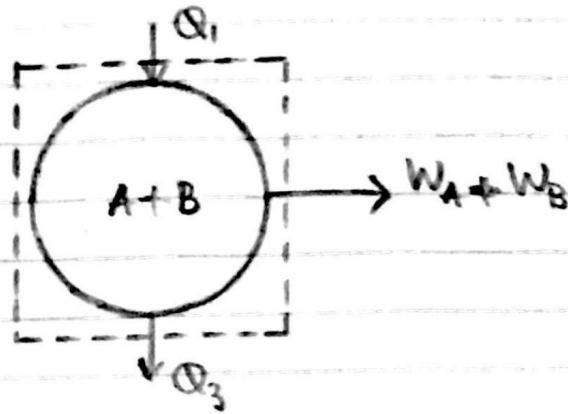
Thanh Nguyen

CHE 201 - HW 14

1.) Sketch



A and B in series



Overall cycle

### Engineering Model

1. All the systems are closed
2.  $Q_{out}$  from A is the same as  $Q_{in}$  into B :  $Q_2$

### Analysis

$$\eta_A = \frac{W_A}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\eta_B = \frac{W_B}{Q_2} = \frac{Q_2 - Q_3}{Q_2} = 1 - \frac{Q_3}{Q_2}$$

$$\eta_{AB} = \frac{\Sigma W}{Q_1} = \frac{W_A + W_B}{Q_1} = \frac{W_A}{Q_1} + \frac{W_B}{Q_1} = \eta_A + \frac{W_B}{Q_2} \cdot \frac{Q_2}{Q_1} = \eta_A + \eta_B (1 - \eta_A)$$

$$= \eta_A + \eta_B - \eta_A \eta_B$$

1 Question 1 20 / 20

✓ - 0 pts Correct

- 5 pts "...in terms of their individual thermal efficiencies"

## 2.) Sketch

Initial state (1)

$$P_1 = 100 \text{ lbf/in}^2$$

$$v_1 = 3.704 \text{ ft}^3/\text{lb}$$

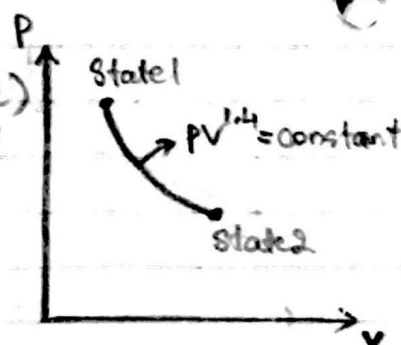
$$T_1 = 1000^\circ\text{R}$$



Final state (2)

$$P_2 = 50 \text{ lbf/in}^2$$

$$v_2, T_2$$



## Engineering Model

1. The system is closed and the mass of gas is  $m$ .
2.  $Pv^{1.4} = \text{constant}$
3.  $\Delta KE = \Delta PE = Q = 0$
4. Work is only in the form of volume expansion
5.  $\Delta u = (0.171)(T_2 - T_1)$

## Analysis

$$P_1 v_1^{1.4} = P_2 v_2^{1.4} \rightarrow v_2 = \left( \frac{P_1 v_1^{1.4}}{P_2} \right)^{\frac{1}{1.4}} = \left( \frac{(100 \text{ lbf/in}^2)(3.704 \text{ ft}^3/\text{lb})^{1.4}}{50 \text{ lbf/in}^2} \right)^{\frac{1}{1.4}} = 6.077 \text{ ft}^3/\text{lb}$$

$$P_1 = 100 \text{ lbf/in}^2 \times \left( \frac{12 \text{ in.}}{1 \text{ ft}} \right)^2 = 1.44 \times 10^4 \text{ lbf/ft}^2$$

$$P_2 = 50 \text{ lbf/in}^2 \times \left( \frac{12 \text{ in.}}{1 \text{ ft}} \right)^2 = 7.2 \times 10^3 \text{ lbf/ft}^2$$

$$\begin{aligned} \frac{W}{m} &= \int_{v_1}^{v_2} P dv = \int_{v_1}^{v_2} \frac{\text{const}}{v^{1.4}} dv = \frac{P_2 v_2 - P_1 v_1}{1 - 1.4} \\ &= \frac{(7.2 \times 10^3 \text{ lbf/ft}^2)(6.077 \text{ ft}^3/\text{lb}) - (1.44 \times 10^4 \text{ lbf/ft}^2)(3.704 \text{ ft}^3/\text{lb})}{-0.4} = 2.396 \times 10^4 \text{ lbf.ft/lb} \end{aligned}$$

$$\Delta E = Q - W \rightarrow \Delta KE + \Delta PE + \Delta U = Q - W \rightarrow \Delta U = -W$$

$$\rightarrow \Delta u = \frac{\Delta U}{m} = \frac{-W}{m} = (-2.396 \times 10^4 \text{ lbf.ft/lb}) \times \frac{1 \text{ Btu}}{778.2 \text{ lbf.ft}} = -30.79 \text{ Btu/lb}$$

$$\Delta u = (0.171 \text{ Btu/(R.lb)})(T_2 - T_1)$$

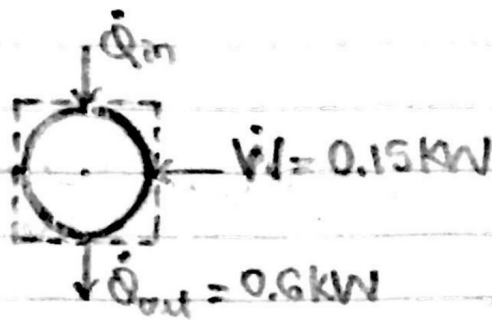
$$\rightarrow T_2 = \frac{\Delta u}{0.171 \text{ Btu/R.lb}} + T_1 = \frac{-30.79 \text{ Btu/lb}}{0.171 \text{ Btu/R.lb}} + 1000^\circ\text{R} = 819.9^\circ\text{R}$$

## 2 Question 2 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong  $V_2$
- 5 pts Wrong work
- 5 pts Wrong internal energy difference
- 5 pts Wrong final temperature
- 20 pts Did not answer 2.53

### 3.) Sketch



#### Engineering Model

1. The system is closed
2. Energy transfers are positive in the directions of arrows

#### Analyses

In a cycle,  $\Delta E = 0 \rightarrow \dot{Q}_{in} + \dot{W} = \dot{Q}_{out}$

$$\rightarrow \dot{Q}_{in} = \dot{Q}_{out} - \dot{W} = 0.6 \text{ kW} - 0.15 \text{ kW} = 0.45 \text{ kW}$$

Coefficient of performance

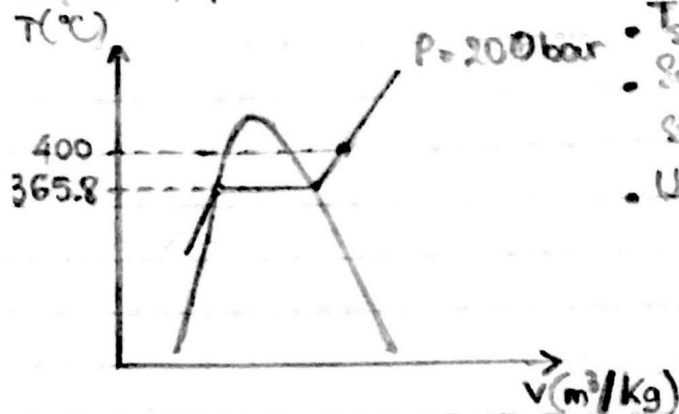
$$\beta = \frac{\dot{Q}_{in}}{\dot{W}} = \frac{0.45 \text{ kW}}{0.15 \text{ kW}} = 3$$

3 Question 3 20 / 20

✓ - 0 pts Correct

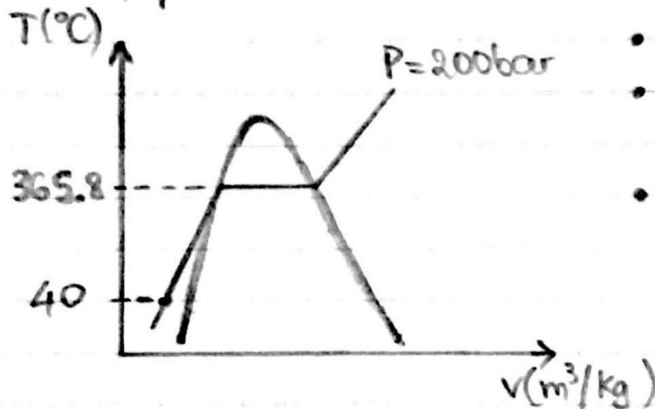
4.)

a.  $T = 400^\circ\text{C}$ ,  $p = 20\text{ MPa} = 200\text{ bar}$



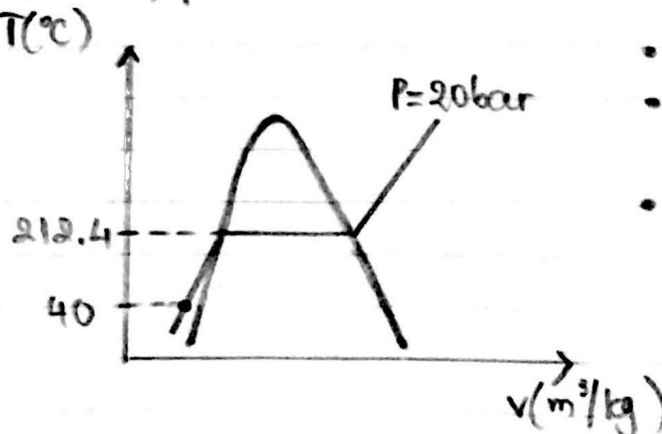
- $T_{\text{sat}} = 365.8^\circ\text{C}$
- Since  $T > T_{\text{sat}}$  at  $p = 200\text{ bar}$ , the state is superheated vapor
- Using table A-4:  
 $v = 0.00994\text{ m}^3/\text{kg}$

b.  $T = 40^\circ\text{C}$ ,  $p = 20\text{ MPa} = 200\text{ bar}$



- $T_{\text{sat}} = 365.8^\circ\text{C}$
- Since  $T < T_{\text{sat}}$  at  $p = 200\text{ bar}$ , the state is compressed liquid
- Using table A-5:  
 $v = 0.9992 \times 10^{-3}\text{ m}^3/\text{kg}$

c.  $T = 40^\circ\text{C}$ ,  $p = 2\text{ MPa} = 20\text{ bar}$



- $T_{\text{sat}} = 212.4^\circ\text{C}$
- Since  $T < T_{\text{sat}}$  at  $p = 20\text{ bar}$ , the state is compressed liquid
- We make assumption that specific volume at the state is approximately equal to that of saturated liquid at the same temperature  $40^\circ\text{C}$ . Using table A-2:

$$v \approx 1.0078 \times 10^{-3}\text{ m}^3/\text{kg}$$

#### 4 Question 4 17 / 20

- 0 pts Correct
- 20 pts Did not answer problem 3.7
- ✓ - 3 pts (a) Wrong sketch of T-v diagram
  - 3 pts (a) Wrong specific volume
  - 3 pts (b) Wrong sketch of T-v diagram
  - 3 pts (b) Wrong specific volume
  - 3 pts (c) Wrong sketch of T-v diagram
  - 3 pts (c) Wrong specific volume

## **The US Phase-Out Schedule for R22 Refrigerants and the Replacement for R22**

### **The US Phase-Out Schedule for R22 and Other HCFC Refrigerants**

The United States Environmental Protection Agency (EPA) released a schedule for phasing out all the HCFC refrigerants as the following list:

- **January 1, 2010:** "Ban on production, import, and use of HCFC-22 and HCFC-142b, except for continuing servicing needs of existing equipment."
- **January 1, 2015:** "Ban on production, import, and use of all HCFCs, except for continuing servicing needs of refrigeration equipment."
- **January 1, 2020:** "Ban on remaining production and import of HCFC-22 and HCFC-142b. After 2020, the servicing of systems with R-22 will rely on recycled or stockpiled quantities."
- **January 1, 2030:** "Ban on remaining production and import of all HCFCs."

EPA implies that homeowners are not required to stop using R-22 air conditioners nor to replace existing equipment. However, in the future, the supplies for R-22 will be limited which causes an increase of the price of maintenance and any other services for R-22. Therefore, it is strongly recommended to replace the R-22 refrigerants completely with non-ozone-depleting refrigerants.

### **The Replacement for R22 Refrigerants**

According to EPA, "the most common alternative to R-22 is R-410A, a non-ozone-depleting HFC refrigerant blend." It is also sold under different names, including GENTRON AZ-20®, SUVA® 410A, and PURON®.

According to Bright Hub Engineering and PETRO, in addition to the R-410A recommended by EPA, there are also some other refrigerants that do not cause the depletion of the ozone layer, including R-134A, R-407A, R-407C, R-507, and R-404A. However, depending on what units the homeowners use, it is advised to use compatible refrigerants top R-22 in order not to cause major failures in systems.

### **References**

1. "Phase Out HCFC Refrigerants to Protect the Ozone Layer." The United States Environmental Protection Agency (EPA), URL:  
[www.epa.gov/sites/production/files/2015-07/documents/phasing\\_out\\_hcfc\\_refrigerants\\_to\\_protect\\_the\\_ozone\\_layer.pdf](http://www.epa.gov/sites/production/files/2015-07/documents/phasing_out_hcfc_refrigerants_to_protect_the_ozone_layer.pdf). Accessed 15 Feb. 2020.
2. "R22 Replacement Options. R22 Alternative Refrigerants." Bright Hub Engineering, URL:  
<https://www.brighthubengineering.com/hvac/62099-alternative-refrigerants-for-r22/>. Accessed 16 Feb. 2020.
3. "R22 (Refrigerant) Replacement." PETRO, URL:  
<https://www.petro.com/cooling/r22-refrigerant-replacement>. Accessed 16 Feb. 2020.

## 5 Question 5 20 / 20

✓ - 0 pts Correct

- 20 pts Did not answer 3.4D
- 5 pts Insufficient component about the phase-out schedule
- 2.5 pts Insufficient component about the phase-out schedule
- 5 pts Insufficient component about the replacer
- 2.5 pts Insufficient component about the replacer
- 1 pts No enough references